

ASSIGNMENT NO.4: Working with Multiple Tables and Joins

Questions: Q1) Create a MySQL database named SeniorCollegeDB and create the following tables: Student Table Column Name Data Type StudentID INT Constraint StudentName VARCHAR(50) PRIMARY KEY Column Name Data Type Constraint City VARCHAR(30) Course Table Column Name Data Type Constraint CourseID INT PRIMARY KEY CourseName VARCHAR(50) Fees DECIMAL(10,2) Enrollment Table Column Name Data Type Constraint EnrollmentID INT PRIMARY KEY StudentID INT FOREIGN KEY CourseID INT FOREIGN KEY EnrollmentDate DATE

ANS:

-- Create Database

```
CREATE DATABASE SeniorCollegeDB;
```

```
USE SeniorCollegeDB;
```

-- Create Student Table

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    City VARCHAR(30)  
);
```

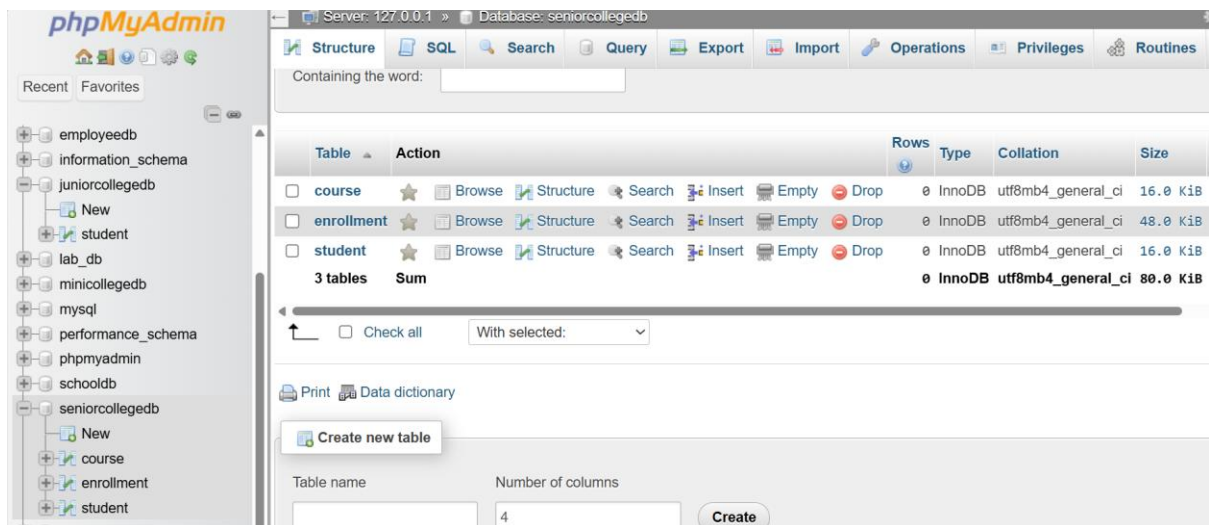
-- Create Course Table

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50),  
    Fees DECIMAL(10,2)  
);
```

-- Create Enrollment Table

```
CREATE TABLE Enrollment (  
    EnrollmentID INT PRIMARY KEY,  
    StudentID INT FOREIGN KEY,  
    CourseID INT FOREIGN KEY,  
    EnrollmentDate DATE
```

EnrollmentID INT PRIMARY KEY,
 StudentID INT,
 CourseID INT,
 EnrollmentDate DATE,
 FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
 FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);



Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table. Make sure:

Requirement
At least one student should not have enrollment
At least one course should not have any enrolled student

ANS:

-- Insert into Student table

INSERT INTO Student VALUES

(1, 'Alice', 'NY'),

(2, 'Bob', 'LA'),

(3, 'Charlie', 'SF'),

(4, 'Dave', 'CHI'),

(5, 'Eve', 'MIA'),

(6, 'Frank', 'SEA');

-- Insert into Course table

INSERT INTO Course VALUES

(101, 'Math', 500.00),

(102, 'Science', 600.00),

(103, 'Art', 300.00),

(104, 'History', 400.00);

-- Insert into Enrollment table

INSERT INTO Enrollment VALUES

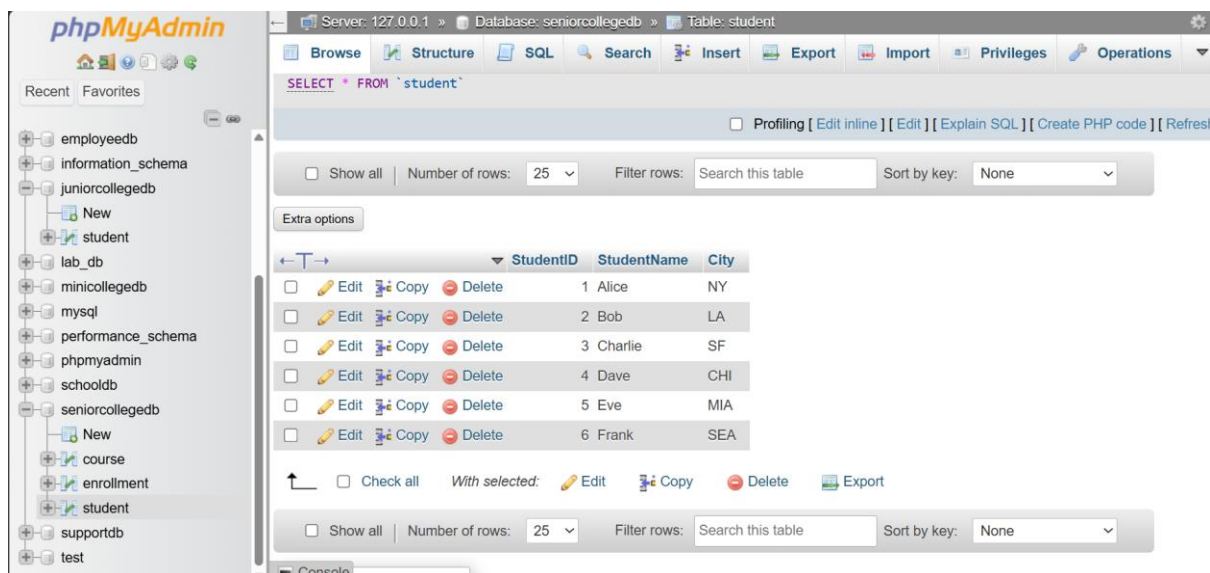
(1001, 1, 101, '2023-01-01'),

(1002, 2, 101, '2023-01-02'),

(1003, 3, 102, '2023-01-03'),

(1004, 4, 103, '2023-01-04'),

(1005, 5, 103, '2023-01-05');



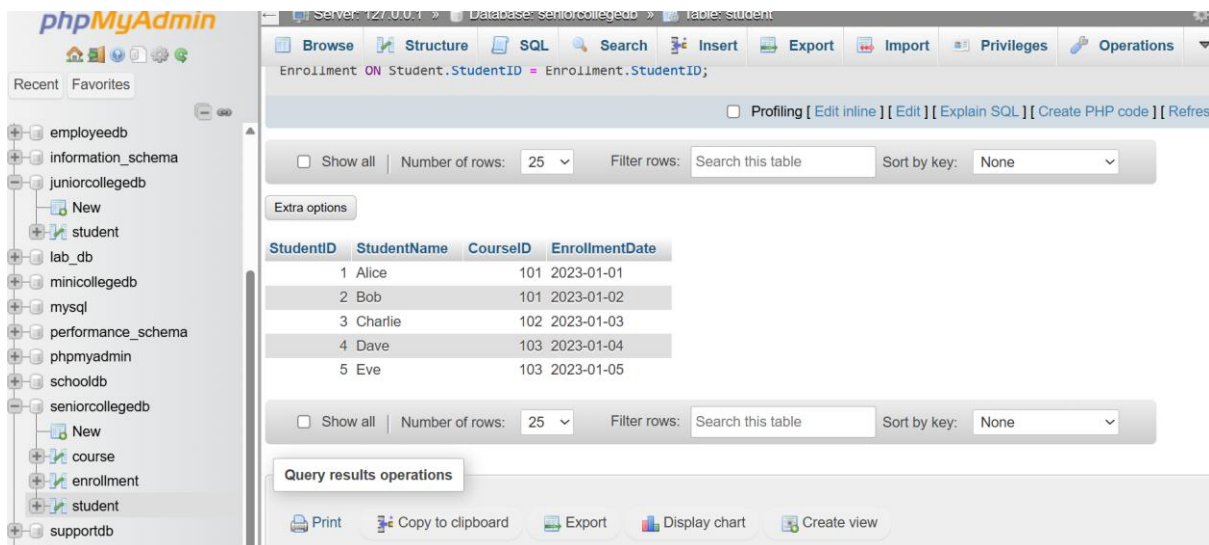
Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses. The output should include:

Column
StudentID

StudentName
CourseID
EnrollmentDate

ANS:

```
SELECT Student.StudentID,
       Student.StudentName,
       Enrollment.CourseID,
       Enrollment.EnrollmentDate
FROM Student
INNER JOIN Enrollment ON
Student.StudentID =
Enrollment.StudentID;
```



StudentID	StudentName	CourseID	EnrollmentDate
1	Alice	101	2023-01-01
2	Bob	101	2023-01-02
3	Charlie	102	2023-01-03
4	Dave	103	2023-01-04
5	Eve	103	2023-01-05

Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables. Use:

Tables
Required
Student
Course
Enrollment

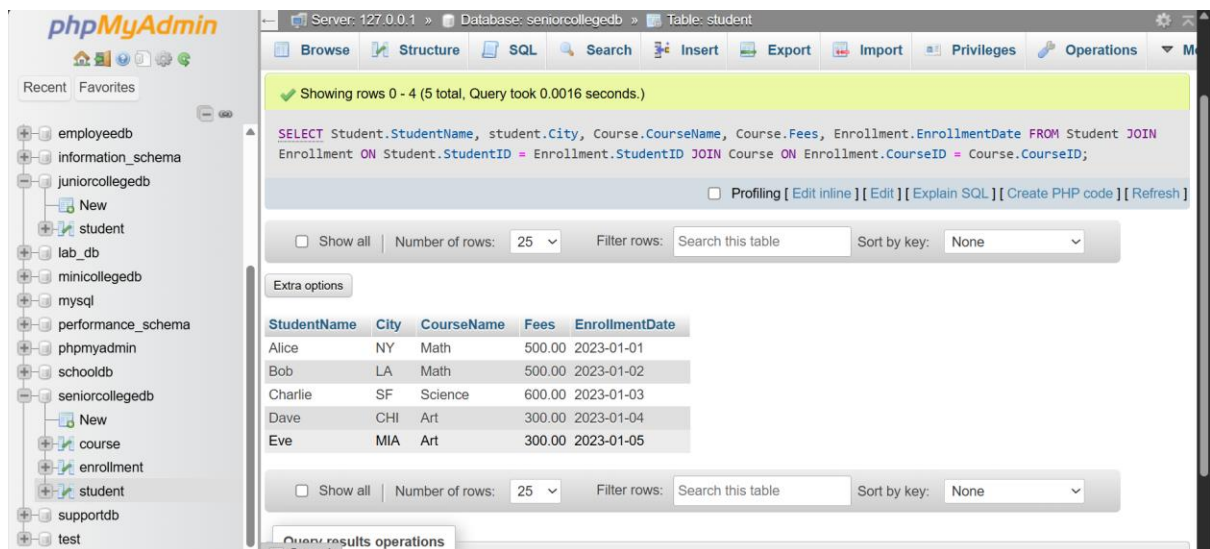
ANS:

```
SELECT Student.StudentName,
```

```

student.City,
Course.CourseName,
Course.Fees,
Enrollment.EnrollmentDate
FROM Student
JOIN Enrollment ON
        Student.StudentID =
        Enrollment.StudentID
JOIN Course ON Enrollment.CourseID
        = Course.CourseID;

```



Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course. The output should include:

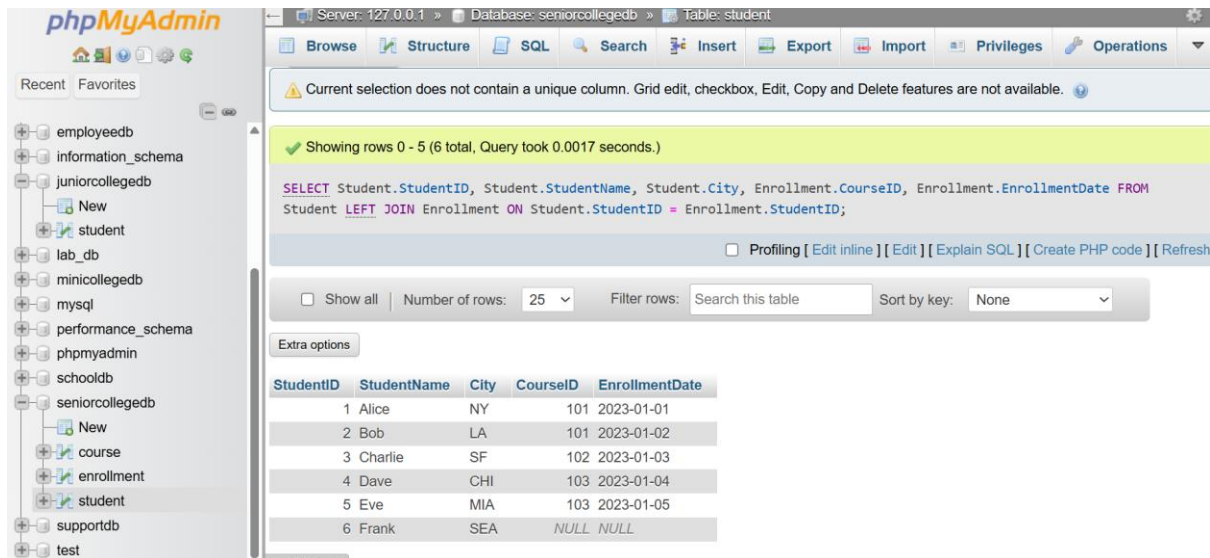
Column
StudentID
StudentName
City
CourseID
EnrollmentDate

ANS:

```

SELECT Student.StudentID,
Student.StudentName,
Student.City,
Enrollment.CourseID,
Enrollment.EnrollmentDate
FROM
    Student
LEFT JOIN Enrollment
ON Student.StudentID = Enrollment.StudentID;

```



Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students. The output should include:

Column
StudentName
CourseID
CourseName
Fees

ANS:

```

SELECT Student.StudentName,
    Course.CourseID,
    Course.CourseName,

```

Course.Fees

FROM Student

JOIN Enrollment ON

Student.StudentID = Enrollment.StudentID

RIGHT JOIN Course ON

Enrollment.CourseID = Course.CourseID;

The screenshot shows the phpMyAdmin interface with the following details:

- Server:** 127.0.0.1
- Database:** seniorcollegedb
- Table:** student
- Query:**

```
SELECT Student.StudentName, Course.CourseID, Course.CourseName, Course.Fees FROM Student JOIN Enrollment ON Student.StudentID = Enrollment.StudentID RIGHT JOIN Course ON Enrollment.CourseID = Course.CourseID;
```
- Result:** Showing rows 0 - 5 (6 total, Query took 0.0010 seconds.)
- Table Data:**

StudentName	CourseID	CourseName	Fees
Alice	101	Math	500.00
Bob	101	Math	500.00
Charlie	102	Science	600.00
Dave	103	Art	300.00
Eve	103	Art	300.00
NULL	104	History	400.00